

Transmission Media

1. Introduction

Transmission media refers to the **communication channel used to transmit data from one device to another in a network**. It is the physical path between a sender and receiver that carries signals such as electrical, optical, or electromagnetic waves.

In computer networks, transmission media plays a very important role because it determines the **speed, reliability, cost, and efficiency of data communication**.

For example, when we send an email, watch a video online, or transfer files between computers, the data travels through different types of transmission media.

Transmission media can be divided into two main categories:

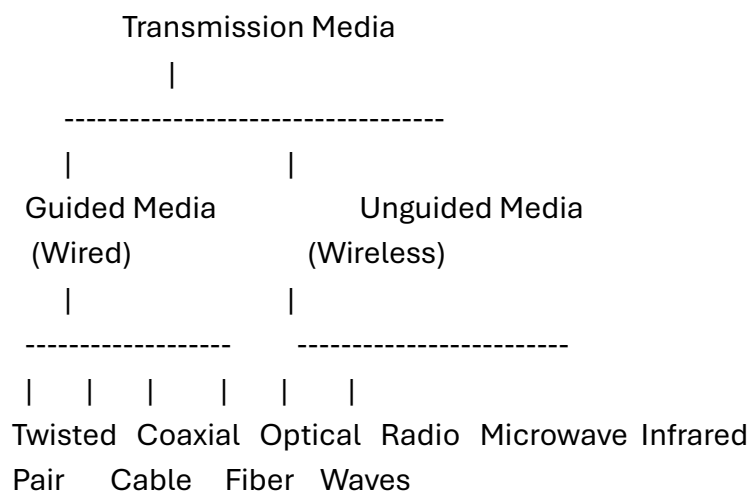
1. **Guided Transmission Media (Wired Communication)**
2. **Unguided Transmission Media (Wireless Communication)**

These two types differ in how signals travel between devices.

2. Types of Transmission Media

Flowchart of Transmission Media

You can draw this in Word using **Insert → Shapes → Flowchart boxes**.



3. Guided Transmission Media (Wired)

Guided media is also called **wired communication** because signals travel through a **physical cable or wire**.

Characteristics

- Uses physical path for communication
- More secure than wireless
- Less interference
- Higher reliability

Guided media is divided into three main types.

3.1 Twisted Pair Cable

Twisted pair cable consists of **two insulated copper wires twisted together** to reduce electromagnetic interference.

It is the **most commonly used transmission media in local area networks (LANs)**.

Types of Twisted Pair Cable

1. UTP (Unshielded Twisted Pair)

- No protective shield
- Cheaper and widely used
- Used in Ethernet networks

2. STP (Shielded Twisted Pair)

- Contains metallic shielding
- Better protection from interference
- More expensive

Advantages

- Low cost
- Easy installation
- Flexible

Disadvantages

- Limited bandwidth
- Short distance communication
- More noise compared to fiber

Applications

- Telephone lines
 - LAN networks
 - Internet connections
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3.2 Coaxial Cable

A coaxial cable contains a **central copper conductor surrounded by insulation, shielding, and an outer cover.**

It provides **better protection against interference than twisted pair cable.**

Structure of Coaxial Cable

It consists of four layers:

1. Inner conductor
2. Insulating layer
3. Metallic shield
4. Outer plastic cover

Advantages

- Higher bandwidth than twisted pair
- Better noise resistance
- More durable

Disadvantages

- Expensive compared to twisted pair
- Harder to install
- Bulky

Applications

- Cable TV networks
 - Internet connections
 - CCTV systems
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3.3 Optical Fiber Cable

Optical fiber is the **most advanced transmission medium**. It transmits data using **light signals instead of electrical signals**.

The cable consists of **very thin strands of glass or plastic called fibers**.

Structure of Optical Fiber

1. Core
2. Cladding
3. Protective coating

Types of Optical Fiber

1. **Single Mode Fiber**
 - Used for long distance communication
 - High speed transmission
2. **Multimode Fiber**
 - Used for shorter distances
 - Lower cost

Advantages

- Very high speed
- Long distance communication
- No electromagnetic interference
- Very high bandwidth

Disadvantages

- Expensive
- Difficult installation
- Fragile

Applications

- Internet backbone
- Telecommunication systems
- Medical equipment
- High speed networks

4. Unguided Transmission Media (Wireless)

Unguided media is also called **wireless communication** because signals travel through **air or space using electromagnetic waves**.

There is **no physical connection between sender and receiver**.

Characteristics

- Uses radio signals
- Easy to install
- Supports mobility
- Covers large areas

Unguided media is divided into several types.

4.1 Radio Waves

Radio waves are widely used for **wireless communication**.

They can travel long distances and can pass through buildings.

Advantages

- Long range communication
- Easy transmission
- No alignment required

Applications

- Radio broadcasting
 - Television broadcasting
 - Mobile communication
 - Wi-Fi networks
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4.2 Microwave Communication

Microwaves are **high frequency electromagnetic waves** used for long distance communication.

Microwave communication requires **line-of-sight transmission**, meaning there must be no obstacle between transmitter and receiver.

Advantages

- High bandwidth
- Fast data transmission
- Suitable for long distances

Disadvantages

- Requires alignment
- Affected by weather conditions

Applications

- Satellite communication
 - Mobile networks
 - Radar systems
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4.3 Infrared Communication

Infrared communication uses **infrared light waves** to transmit data over short distances.

It is commonly used in **remote controls and short-range devices**.

Advantages

- Secure communication
- Low interference
- Cheap technology

Disadvantages

- Very short range
- Cannot pass through walls

Applications

- TV remote controls
- Wireless keyboards
- Short distance communication devices

5. Difference Between Guided and Unguided Media

Feature	Guided Media	Unguided Media
Medium	Physical cables	Air or space
Speed	Generally higher	Depends on signal strength
Security	More secure	Less secure
Cost	Installation cost higher	Lower installation cost
Examples	Twisted pair, Coaxial, Fiber Radio, Microwave, Infrared	

6. Conclusion

Transmission media is an essential component of **computer networks and communication systems**. It provides the path through which data travels from one device to another.

Guided media such as **twisted pair, coaxial cable, and optical fiber** offer secure and reliable communication through physical cables. Unguided media such as **radio waves, microwaves, and infrared** allow wireless communication over long distances.

With the advancement of technology, **optical fiber and wireless communication systems** are becoming increasingly important for high-speed data transmission and global connectivity.